

FIN 405 - Portfolio Allocation and Asset Pricing

Professor: Prof. James Morrison | jmorrison@university.edu | Business School 215

Office Hours: Tue & Thu 3:30-5:00 PM, or by appointment

Term: Fall 2026 | Credits: 3

Location: See schedule below

Course Description

This advanced finance course examines the theory and practice of portfolio construction and asset pricing. Students will study Modern Portfolio Theory (MPT), the Capital Asset Pricing Model (CAPM), multi-factor models (Fama-French), options pricing (Black-Scholes), and risk management frameworks. Emphasis is placed on empirical implementation: students will use Python (pandas, yfinance, statsmodels) to download real market data and test theoretical predictions. The course meets Tuesday and Thursday from 2:00-3:15 PM in Business School 305.

Learning Objectives

Students completing this course will:

1. Construct mean-variance efficient portfolios and the efficient frontier.
2. Apply CAPM and Fama-French 3-factor model to equity returns.
3. Price European options using Black-Scholes and binomial trees.
4. Measure and manage portfolio risk (VaR, CVaR, beta hedging).
5. Present original empirical findings in a professional research report.

Required Materials

Primary: 'Investments' by Bodie, Kane, and Marcus, 12th ed. (McGraw-Hill).

Supplement: 'Options, Futures, and Other Derivatives' by Hull, 11th ed.

Software: Python 3.11+, Jupyter, pandas, numpy, scipy, yfinance, statsmodels.

Bloomberg terminal access available in the Finance Lab (Business School B102).

Assignments & Due Dates

Case Study 1 - Modern Portfolio Theory - Due: September 25, 2026 (10%)

Download 5 years of daily returns for 10 S&P 500 stocks using yfinance. Compute the covariance matrix, construct the minimum-variance portfolio and the tangency portfolio, and plot the efficient frontier. Submit a Jupyter notebook and a 2-page write-up. Due by 11:59 PM on Gradescope.

Problem Set 1 - CAPM and Beta Estimation - Due: October 9, 2026 (8%)

Theoretical problems plus empirical exercise: estimate betas for 5 stocks via OLS regression against the market index. Test the CAPM prediction using the Fama-MacBeth cross-sectional regression. Submit PDF on Gradescope.

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Midterm Exam - Due: October 27, 2026 (22%)

In-class exam on MPT, CAPM, and multi-factor models. Covers Chapters 1-10 of Bodie et al. Closed-book; one formula sheet (one-sided, 8.5x11) allowed. Financial calculator permitted. Held 2:00-3:15 PM in Business School 305.

Portfolio Project - Proposal - Due: November 6, 2026 (5%)

One-page proposal describing your portfolio strategy: asset universe (minimum 15 securities across 3 sectors), time period, pricing model, and risk targets. Include a preliminary data source plan. Submit on course portal.

Case Study 2 - Options Pricing and the Volatility Surface - Due: November 20, 2026 (10%)

Price a set of European put and call options on SPY using Black-Scholes. Compute implied volatilities, plot the volatility surface, and discuss deviations from BSM assumptions. Submit Jupyter notebook + 3-page report.

Portfolio Project - Final Report - Due: December 7, 2026 (25%)

Full empirical portfolio analysis (8-10 pages + appendices). Must include: factor model estimation, portfolio optimization, out-of-sample back-test (rolling 6-month window), Sharpe ratio and drawdown analysis, and risk decomposition. Jupyter notebook required. Groups of 1-3 students.

Final Exam - Due: December 16, 2026 (20%)

Comprehensive 2-hour final covering all material including options pricing and risk management. Held in registrar-assigned room. Formula sheet allowed (two-sided, 8.5x11). Financial calculator permitted.

Course Schedule Notes

Week 1 (Sep 8-10): Risk and return basics, portfolio math
Week 2 (Sep 15-17): Modern Portfolio Theory, efficient frontier
Week 3 (Sep 22-24): CAPM derivation and assumptions
Week 4 (Sep 29-Oct 1): CAPM empirical tests, beta estimation
Week 5 (Oct 6-8): Fama-French 3 & 5 factor models
Week 6 (Oct 13-15): Momentum, liquidity, and anomalies
Week 7 (Oct 20-22): Review; MIDTERM EXAM Oct 27
Week 8 (Oct 27-29): Fixed income basics, duration and convexity
Week 9 (Nov 3-5): Options: payoffs, put-call parity
Week 10 (Nov 10-12): Black-Scholes model, Greeks
Week 11 (Nov 17-19): Volatility surface, exotic options intro
Week 12 (Nov 24): Value at Risk and CVaR [Thursday only - no Tuesday class]
Week 13 (Dec 1-3): Portfolio risk management, tail risk
Week 14 (Dec 8-10): Student project presentations + review
FINAL EXAM: December 16, 2026

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Grading Policy

Case Studies (2): 20%

Problem Set 1: 8%

Portfolio Project: 30% (proposal 5% + final 25%)

Midterm Exam: 22%

Final Exam: 20%

Late policy: Case studies and problem sets lose 15% per day. Project proposal late submissions not accepted. Grade scale: A 93+, A- 90, B+ 87, B 83, B- 80, C+ 77, C 73, D 60, F <60.